Aaryaa Moharir
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EDUCATION

Carnegie Mellon University, Pittsburgh, PA *May 2028*
Master's of Artificial Intelligence Engineering – Information Security
Coursework: Cyber Risk Management, AI Applications of Information Security, Systems and Tool Chains for AI Engineers
University of Texas at Dallas, Richardson, TX *May 2026*
Bachelor of Computer Science *GPA: 4.0/4.0*
Coursework: Cyber Defense Lab, Database Systems, Adv. Algorithms, Data Structures & Algorithms, Operating Systems

SKILLS

Languages: Java, Python, C#, C++ | **Security:** MITRE ATT&CK, Penetration Testing, Malware Analysis, Data Security, CTFs, OWASP Top 10
Tools & Technologies: Wireshark, Metasploit, MongoDB, SQL, Linux, AWS, Google Cloud, Apache Kafka, MapReduce, PySpark | **Development:** React, JavaScript, REST APIs, Database Design

EXPERIENCE

Security Engineering Intern – Amazon Web Services June 2026 – August 2026
- Designed a security framework for Agentic AI workflows across SDO and AWS, modeling and mitigating 32 security threats.
- Authored testing playbooks, threat model templates, and conducted security reviews using the framework.
- Engineered agent scripts and tools using Python per threat for automated vulnerability detection.
- Worked with scientists to implement an internal benchmarking framework to systematically evaluate agentic AI skills, defining evaluation datasets, performance metrics, and automated test pipelines to measure precision, recall, accuracy, and f1 score.

Cybersecurity Research Intern – DNIF Hypercloud June 2025 – August 2025
- Built a **Detection Use Case Matrix** and **Threat Hunting Playbook**, authoring pseudo-Sigma detection logic across event IDs and log sources to flag process hollowing, LSASS access, and HTTP-based exfiltration.
- Delivered a **Detection Coverage Map** identifying visibility gaps across 8 use cases and presenting remediation recommendations.

Undergraduate Researcher in the Intelligent Systems Lab – UT Dallas May 2024 – December 2025
- Implemented **gradient-based, semantic-preserving input adaptation techniques** in Python to refine images for uncertain classifications within 6 pretrained ResNet models, improving model performance through contrast adjustment, rotation, and validation.
- Contributed to research on GPU bugs in tile programs, analyzing **900+ samples** to map root causes to symptoms, and developed a **structured framework** for automated bug detection and repair, including the preliminary tool **Tile-CBDetect**.

Undergraduate Researcher in the System Security Lab – UT Dallas August 2024 – May 2025
- Contributed to the development of attack pathway detection techniques by analyzing network traffic such as file access, IP connections, and process activity to identify malicious behavior patterns and implemented 6 tracking strategies using C#.
- Built a database system to aggregate and query event counts using MongoDB, SQL, and C#, reducing data query time by **30%** for large-scale network analysis.
- Integrated and adapted the MAGIC anomaly detection model for threat identification, customizing the pipeline for compatibility with the lab's proprietary dataset.

PROJECTS

Developer - YouTravel | An AI-driven travel analytics platform January 2025 – May 2026
- Engineered an AI travel platform that extracts YouTube video transcripts, extracts trending narratives and their embeddings, and utilizes **HDBSCAN** clustering to generate narrative insights on trending destinations
- Designed an indexed MongoDB backend and containerized dependencies with Docker.
- Automated scheduled data workflows by deploying containerized microservices to Google Cloud Run, utilizing Artifact Registry and a robust GitHub CI/CD pipeline for hands-off weekly clustering execution.

Developer - ResearchMate | An Agentic Solution to Understand Research Papers Better September 2025 – May 2026
- Built **a RAG-based** platform to process uploaded academic papers, generate summaries and deliver contextual chatbot responses.
- Developed an **MCP server** with custom Python tool integrations (Google Search, arXiv APIs) to dynamically inject external knowledge sources into LLM prompts, reducing hallucination rates.
- Utilized Qdrant for vector embeddings and built FastAPI endpoints connecting the React/Tailwind frontend to the Python backend.

Developer – ECHO 911 | Winner (1st place), Breaking Barriers Hackathon (AWS, AT&T, Deloitte) September 2025- September 2025
- Built a **real-time dispatcher dashboard** using **React** (frontend) and **Python** (backend) to display 911 calls transcripts, display policies to follow on scene, and support dispatcher decision-making.
- Leveraged **AWS services** (Connect, Lambda, Bedrock, Strands, S3, Polly) to transcribe calls, extract key details, and evaluate incident severity.
- Integrated **OpenStreetMap API, Open-Meteo API, Amazon Bedrock, and Claude Sonnet** to enrich emergency location data and automate first responder mapping.

Detect and Defend – Team Lead December 2023 – May 2024
- Led and mentored a 5-person team developing a Python-based machine learning model to attack Federated Learning systems by reconstructing original training datasets through gradient reversal techniques.
- Utilized LeNet-5 architecture with multiple datasets including MNIST to train and evaluate a General Regression Neural Network model.
- Demonstrated security vulnerabilities in distributed learning systems and proposed defensive countermeasures.

ACCOMPLISHMENTS & CERTIFICATES

- 1st place out of 36 teams at the Breaking Barriers Hackathon hosted by AWS, AT&T, and Deloitte September 2025
- Published "Characterizing Real-World Bugs in Tile Programs for Automated Bug Detection" May 2025